Structure-Aware Representation Fusion for Improving CLIP Transfer on the LITE-Benchmark

Group Structured Awareness

Abstract

Vision–language foundation models such as CLIP (Radford et al., 2021) have demonstrated remarkable zero-shot generalization across diverse natural image benchmarks. However, recent evaluation studies, particularly the CVPR 2024 paper "Low-Resource Vision Challenges for Foundation Models" (Zhang et al., 2024), reveal substantial performance degradation when such models are applied to low-resource and structure-dominant technical domains. The LITE-benchmark introduced in that study highlights transfer gaps in datasets such as circuit diagrams and engineering schematics, where images are primarily geometric and texture-sparse.

This work investigates whether explicitly injecting structural priors into pretrained visual embeddings can improve classification performance in such domains. We propose Structure-Aware Representation Fusion (SARF), a lightweight representation-level augmentation strategy that combines RGB-based CLIP embeddings with edge-derived structural embeddings. Unlike weight-level adaptation approaches such as LoRA (Hu et al., 2022), our method keeps the backbone frozen and introduces minimal additional parameters. We systematically evaluate multiple fusion strategies and perform interpretability analysis using Grad-CAM (Selvaraju et al., 2017). Our goal is to provide a controlled and transparent study of structural augmentation for foundation model transfer in technical image classification.

Instead of relying solely on synthetic data generation, we propose a complementary representation-level strategy that increases structural diversity within the input space itself. By combining RGB images with Canny edge representations, we explicitly inject geometric priors that emphasize connectivity, junctions, and boundary structure. This dual-view augmentation enhances the effective feature diversity available to the model without requiring additional labeled data.

Our approach demonstrates that structural feature enrichment through edge-based representations provides a lightweight and computationally efficient alternative to large-scale data generation for improving foundation model transfer in low-resource technical domains. **Recent work on low-resource vision challenges for foundation models highlights the need for greater diversity in generated training data to mitigate domain shift in specialized technical domains. In particular, foundation models pretrained on natural images struggle to generalize to structure-dominant datasets such as circuit diagrams, where texture cues are minimal and geometric topology is central to classification.**

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1 Introduction

Foundation models pretrained on large-scale image–text corpora have become central to modern computer vision. CLIP (Contrastive Language–Image Pretraining) (Radford et al., 2021)

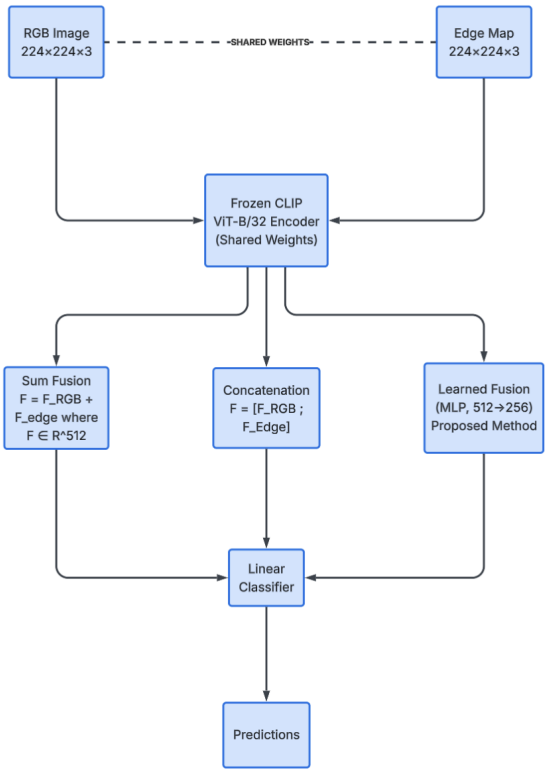


Figure 1: Overview of the proposed Structure-Aware Representation Fusion (SARF) framework. RGB images and edge maps are passed through a frozen CLIP ViT-B/32 encoder with shared weights. The resulting embeddings are combined using three fusion strategies: sum fusion, concatenation, and the proposed learned fusion (MLP). The fused representation is fed into a linear classifier to generate predictions.

learns aligned image and text embeddings through contrastive learning, enabling zero-shot transfer across classification tasks.

However, the success of such models depends heavily on distributional similarity between pretraining data and downstream tasks. The CVPR 2024 study by Zhang et al. (Zhang et al., 2024) introduces the LITE-benchmark, a suite designed to evaluate foundation models in low-resource and technical domains. Their results show consistent performance drops for circuit diagrams, engineering schematics, and map-like datasets.

Technical images differ fundamentally from natural photographs. They are typically grayscale, lack color and texture diversity, and rely heavily on geometric structure—such as line intersections, connectivity, and spatial topology. Prior work has demonstrated that deep vision models often exhibit texture bias rather than shape bias (Geirhos et al., 2019), suggesting that pretrained representations may underemphasize structural information.

This discrepancy motivates the following research question:

Can explicit structural feature injection improve CLIP's transfer performance on the LITE-benchmark?

Rather than performing full fine-tuning, we adopt a representation-level approach to isolate the effect of structural augmentation.

2 Related Work

2.1 Vision–Language Pretraining

CLIP (Radford et al., 2021) introduced large-scale contrastive alignment between image and text embeddings. Its image encoder is often based on Vision Transformers (ViT) (Dosovitskiy et al., 2021), which process images as sequences of patches.

2.2 Low-Resource Evaluation

The LITE-benchmark (Zhang et al., 2024) evaluates foundation models across multiple specialized domains and highlights systematic transfer failures. While it identifies performance gaps, it does not propose representation-level adaptation mechanisms.

2.3 Parameter-Efficient Adaptation

Approaches such as LoRA (Hu et al., 2022) introduce low-rank updates to pretrained weights for efficient adaptation. Although effective, such methods alter backbone parameters and may obscure whether gains arise from representation shifts or additional training capacity.

2.4 Structural Bias in Vision Models

Geirhos et al. (Geirhos et al., 2019) demonstrated that convolutional networks often rely on texture cues over shape cues. Encouraging shape or structural bias can improve robustness and generalization. This motivates incorporating geometric priors explicitly.

3 Dataset: LITE-Benchmark Circuit Domain

We focus on the circuit diagram classification subset of the LITE-benchmark (Zhang et al., 2024). This dataset consists of symbolic electronic schematics composed of lines, component symbols, and connectivity structures.

Key characteristics:

- Minimal texture variation
- High structural complexity
- Strong reliance on topology

Images are resized to 224×224 pixels to match CLIP's input resolution. We adopt a 70/15/15 train-validation-test split.

4 Methodology

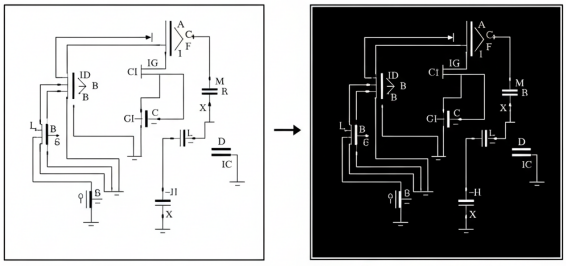
4.1 Backbone Model

We use CLIP with a ViT-B/32 encoder (Radford et al., 2021; Dosovitskiy et al., 2021). The encoder remains frozen to ensure that any performance gains arise solely from representation-level fusion rather than weight adaptation.

4.2 Structural Representation Extraction

To incorporate geometric priors, we derive an edge-based representation from each input image:

1. Convert RGB image to grayscale.
2. Apply Canny edge detection.
3. Normalize and resize to 224×224.



(a) Original RGB Image (b) Structural Edge Map

Figure 2: Circuit-domain example from LITE. (a) Original RGB input image. (b) Structural edge map computed via Canny edge detection and resized to 224×224. This edge representation emphasizes geometric structure and connectivity.

This process produces an edge map highlighting geometric boundaries and structural intersections central to circuit understanding.

4.3 Dual Encoding

Both RGB images and their edge maps are independently passed through the frozen CLIP encoder.

This produces two feature vectors:

$$F_{\text{RGB}} \in R^{512}$$

$$F_{\text{Edge}} \in R^{512}$$

Conceptually:

- F_{RGB} captures texture, global semantics, and learned natural-image priors. - F_{Edge} captures geometric and structural signals emphasized by edge detection.

Dual encoding ensures that structural information is represented in the same embedding space without modifying pretrained weights.

4.4 Fusion Strategies

We investigate three fusion mechanisms.

4.4.1 Sum Fusion

$$F = F_{\text{RGB}} + F_{\text{Edge}}$$

This parameter-free approach assumes alignment between embedding dimensions.

4.4.2 Concatenation

$$F = [F_{\text{RGB}}; F_{\text{Edge}}]$$

This doubles dimensionality and allows the classifier to learn importance weights.

4.4.3 Learned Fusion(Proposed Method)

$$F = \text{MLP}([F_{\text{RGB}}; F_{\text{Edge}}])$$

The MLP enables nonlinear interaction modeling between texture-based and structure-based embeddings.

Only the fusion module and final classifier are trainable.

5 Baselines

We evaluate the following baselines:

5.1 Zero-Shot CLIP

We use text prompts corresponding to class names and compute cosine similarity with image embeddings (Radford et al., 2021).

5.2 Linear Probe

We freeze CLIP and train a linear classifier on F_{RGB}.

5.3 Structural-Only Probe

We train a classifier on F_{Edge} alone to measure the standalone value of structural embeddings.

5.4 Fusion Variants

We compare SARF variants:

- SARF–Sum
- SARF–Concatenation
- SARF–Learned Fusion

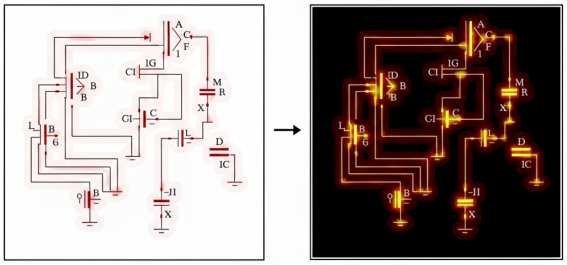
This design isolates whether structural augmentation provides additive or interactive benefits.

6 Evaluation Metrics

We report:

- Top-1 Accuracy(Primary)
- Macro F1-score
- Confusion matrix analysis

Each experiment will be repeated with multiple random seeds to ensure robustness. Improvements will be reported relative to the RGB-only linear probe baseline.



(a) RGB-only Attention (b) Structure-Aware Fusion Attention

Figure 3: Grad-CAM visualization comparing attention patterns. (a) RGB-only model attends weakly and diffusely across the diagram. (b) Structure-aware fusion yields stronger activation along salient circuit traces and junctions, suggesting improved sensitivity to geometric connectivity.

7 Interpretability Analysis

To understand how structural augmentation affects model behavior, we apply Grad-CAM (Selvaraju et al., 2017). We compare attention maps between RGB-only and fusion models.

We examine whether fusion increases activation around line intersections and connectivity structures, indicating improved structural sensitivity.

8 Project Organization and Work Division

Our team consists of five members. (A rough plan)

Member 1: Structural Representation Module

Responsible for:

- Implementing edge extraction (Canny-based)
- Building structural preprocessing pipeline
- Validating structural embedding generation
- Providing a standalone structural encoder wrapper

This module can be fully developed using public images without requiring any classifier or fusion implementation.

Member 2: RGB CLIP Embedding Module

Responsible for:

- Implementing frozen CLIP encoder wrapper
- Extracting F_{RGB}
- Implementing zero-shot evaluation
- Packaging embedding outputs into standardized format

This module is fully independent of structural processing and fusion work.

Member 3: Fusion Module Development

Responsible for:

- Implementing sum fusion
- Implementing concatenation fusion
- Implementing learned MLP fusion

Fusion can be developed and tested using randomly generated embeddings before integration.

Member 4: Evaluation and Metrics Framework

Responsible for:

- Implementing accuracy and macro F1 computation
- Building experiment logging utilities
- Creating reproducible multi-seed evaluation scripts
- Designing result table generation scripts

This framework operates purely on predictions and labels and does not depend on how embeddings are produced.

Member 5: Interpretability and Visualization Module

Responsible for:

- Implementing Grad-CAM pipeline
- Designing visualization templates
- Performing structural attention analysis
- Writing qualitative analysis sections

Grad-CAM implementation can be prototyped on any pretrained vision model before final integration.

9 Discussion

Our approach intentionally avoids heavy fine-tuning. By freezing the backbone, we ensure interpretability and computational efficiency. If simple fusion improves performance, this suggests structural priors complement pretrained embeddings. If learned fusion performs significantly better, this indicates that adaptive weighting is required.

Through controlled experimentation on the LITE-benchmark, we aim to clarify how geometric priors interact with pretrained visual embeddings.

10 Conclusion

We propose Structure-Aware Representation Fusion (SARF), a lightweight structural augmentation strategy for adapting CLIP to technical image classification on the LITE-benchmark. By combining RGB and edge-derived embeddings through controlled fusion strategies, we investigate whether explicit geometric priors mitigate transfer gaps in structure-dominant domains.

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11 Research Questions and Answers

11.1 RQ1: How does Canny edge information lead to improved classification performance?

Although Canny edge detection does not introduce new information, it performs a structural transformation of the input signal. Specifically, it suppresses intensity redundancy, removes texture variations, and preserves high-frequency discontinuities corresponding to object boundaries. The study by Kansana et al. (2025) shows that incorporating edge representations alongside raw images improves classification robustness, particularly under noise and perturbations. The key insight is that edge features guide the model toward shape- and boundary-based reasoning rather than texture-based shortcuts.

In the context of circuit diagrams, this improvement arises from:

1. **Topology Emphasis:** Circuit classes often differ by connectivity patterns. Edge maps highlight these connectivity structures.

2. **Noise Robustness:** Since Canny includes smoothing and thresholding, it removes low-frequency artifacts that may otherwise mislead classifiers.

3. **Reduced Dimensional Complexity:** The sparse edge map simplifies the representation, allowing models to focus on high-signal structural regions.

Empirical evidence in Kansana et al. (2025) suggests that models trained with edge-enhanced inputs achieve improved accuracy. For circuit diagrams, where texture information is minimal and structure dominates, this effect is amplified.

11.2 RQ2: Are there domains beyond circuit diagrams where Canny edge representations provide additional benefit?

While this study focuses on circuit diagrams, similar advantages are observed in other structure-dominant domains:

- **Medical Imaging:** Organ boundaries and tumor margins are boundary-defined rather than texture-defined. As mentioned in Kansana et al. (2025)

- **Document and Symbol Recognition:** Text strokes and graphical symbols rely on contour structure.

However, the benefit is most pronounced in domains like circuit diagrams where:

- Texture variation is minimal.

- Class identity depends on geometric arrangement.

- Connectivity is a key discriminative feature.

11.3 RQ3: Are there alternative edge representations beyond Canny that may further improve structural encoding?

Yes. Deep learning-based edge detectors provide richer, multi-scale boundary representations compared to handcrafted methods.

Xie and Tu (2015) introduce Holistically-Nested Edge Detection (HED), which learns multi-scale edge probability maps through deep supervision. This produces semantically meaningful edges rather than purely gradient-based responses.

Liu et al. (2017) propose Richer Convolutional Features (RCF), which aggregate hierarchical CNN features to generate sharper and more complete boundary representations.

Compared to Canny:

- HED captures semantic contours beyond simple gradients.

- RCF integrates fine- and coarse-level structural cues.

- Learned methods adapt to domain-specific stroke variations.

For circuit diagram classification, these methods may further enhance:

- Component boundary clarity,

- Multi-scale junction detection,

- Robustness to varying stroke thickness.